

Design and Analysis of Algorithm (DAA), Lab-01

BTech (CS-B and CE), 3rd Semester
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Questions

1. **Put them in Order:** Using implementation in C, place the given functions in a list by *increasing order of growth* for sufficiently large values of n .

$n \log_2 n$	$12\sqrt{n}$	$\frac{1}{n}$	$n^{\log_2 n}$
$100n^2 + 6n$	$n^{0.51}$	$n^2 - 324$	$50n^{0.5}$
$2n^3$	3^n	$2^{32}n$	$\log_2 n$

2. **Fair vs Biased coin:** Using simulation in C, show that the probability of getting a *HEAD* by tossing a *fair coin* is about 0.5. Extend your simulation to compare fair vs biased coin-tossing experiments.

[Interesting Read] Check this article on arXiv titled “*Random Coin Tossing with unknown bias*” for more insight.

3. **Performance analysis of bubble sort:** Using C, implement two different versions of bubble sort simulation for randomised data sequences as follows:

(i) Bubble sort that terminates if the array is sorted before the $(n - 1)^{th}$ pass.

(ii) Bubble sort that always completes the $(n - 1)^{th}$ pass.

Plot the number of comparisons in both cases to analyse their efficiency.

4. **Towers of Hanoi (ToH):** Simulate the solution to the ToH problem using C. Plot the total number of moves required for solving the problem of n -discs. What can you conclude about your algorithm from the plot obtained?
 5. **Find the partition point:** Consider an array A with n elements containing a run of 0's followed by a run of 1's. Implement a method to find out the exact point of transition between them.
 6. **Element uniqueness:** For given n random numbers, implement a method in C to check if there are any duplicates. What can you conclude about your method for a sufficiently large value of n ?
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